1. What is the SI unit of electrical potential difference?

(A) Ampere
(B) Volt
(C) Ohm
(D) Coulomb

2. Ohm's law states that current is directly proportional to voltage at constant:

(A) Pressure
(B) Temperature
(C) Volume
(D) Resistance

3. Kirchhoff's Current Law (KCL) is based on the conservation of:

(A) Energy
(B) Charge
(C) Momentum
(D) Mass

4. Kirchhoff's Voltage Law (KVL) is based on the conservation of:

(A) Energy
(B) Charge
(C) Magnetic flux
(D) Power

5. The reciprocal of electrical resistance is termed:

(A) Conductance
(B) Inductance
(C) Capacitance
(D) Susceptance

6. What is the SI unit of electrical conductance?

(A) Ohm
(B) Siemens (Mho)
(C) Henry
(D) Farad

7. In an ideal voltage source, the internal resistance is:

(A) Zero
(B) Infinite
(C) 1 Ohm
(D) Variable

8. In an ideal current source, the internal resistance is:

(A) Zero
(B) Infinite
(C) 1 Ohm
(D) Negative

9. Two resistors of 6 Ohms and 3 Ohms connected in parallel have an equivalent resistance of:

(A) 9 Ohms
(B) 2 Ohms
(C) 0.5 Ohms
(D) 18 Ohms

10. Three resistors of 10 Ohms each connected in series yield a total resistance of:

(A) 3.33 Ohms
(B) 30 Ohms
(C) 10 Ohms
(D) 100 Ohms

11. Electrical power dissipated in a resistor of resistance R carrying current I is:

(A) V / I
(B) I^2 * R
(C) I / R
(D) V^2 * I

12. If the length of a conductor is doubled while its cross-sectional area remains unchanged, its resistance:

(A) Is halved
(B) Is doubled
(C) Remains identical
(D) Increases 4-fold

13. If the radius of a cylindrical wire is doubled, its electrical resistance:

(A) Doubles
(B) Halves
(C) Reduces to 1/4
(D) Increases 4-fold

14. A junction point in a circuit where three or more branches meet is defined as a:

(A) Loop
(B) Node
(C) Mesh
(D) Branch

15. A closed path in a network which does not contain any other closed path inside it is a:
- **(A)** Mesh
- **(B)** Node
- **(C)** Supernode
- **(D)** Branch

16. Thevenin's theorem reduces an active linear network to an equivalent voltage source Vth in:
- **(A)** Series with Rth
- **(B)** Parallel with Rth
- **(C)** Series with short circuit
- **(D)** Parallel with open circuit

17. Norton's theorem replaces an active linear network with a Norton current source In in:
- **(A)** Series with Rn
- **(B)** Parallel with Rn
- **(C)** Series with Vth
- **(D)** Parallel with zero resistance

18. The Norton current In is equal to the:
- **(A)** Open circuit voltage
- **(B)** Short circuit current across load terminals
- **(C)** Current through internal source
- **(D)** Zero

19. The relationship between Thevenin voltage, Norton current, and Thevenin resistance is:
- **(A)** Vth = In * Rth
- **(B)** In = Vth * Rth
- **(C)** Rth = In / Vth
- **(D)** Vth = In / Rth

20. The Superposition theorem is applicable only to circuits that are:
- **(A)** Non-linear
- **(B)** Linear and bilateral
- **(C)** Unilateral
- **(D)** Time-variant

21. When applying the Superposition theorem, an independent voltage source is deactivated by:
- **(A)** Replacing it with an open circuit
- **(B)** Replacing it with a short circuit (0 V)
- **(C)** Multiplying by zero
- **(D)** Reversing polarity

22. When applying the Superposition theorem, an independent current source is deactivated by:
- **(A)** Replacing it with a short circuit
- **(B)** Replacing it with an open circuit (0 A)
- **(C)** Grounding both terminals
- **(D)** Connecting in series

23. Maximum power is transferred from a source to a resistive load when load resistance RL equals:
- **(A)** Zero
- **(B)** Infinity
- **(C)** Thevenin resistance Rth
- **(D)** Half of Rth

24. What is the maximum power transfer efficiency of a DC circuit delivering maximum power to RL?
- **(A)** 100%
- **(B)** 75%
- **(C)** 50%
- **(D)** 25%

25. In a star-connected network of three identical resistors R, the equivalent delta resistors each equal:
- **(A)** R / 3
- **(B)** R
- **(C)** 3 * R
- **(D)** 9 * R

26. In a delta-connected network of three identical resistors R, the equivalent star resistors each equal:
- **(A)** R / 3
- **(B)** 3 * R
- **(C)** R
- **(D)** R / 9

27. A supernode is formed in nodal analysis when a voltage source is connected between:
- **(A)** A non-reference node and ground
- **(B)** Two non-reference nodes
- **(C)** Two reference nodes
- **(D)** A current source and resistor

28. A supermesh is formed in mesh analysis when a:
- **(A)** Voltage source is shared by two meshes
- **(B)** Current source is shared by two meshes
- **(C)** Resistor is shared by two meshes
- **(D)** Node has four branches

29. In nodal analysis of a circuit containing N essential nodes, the number of independent KCL equations is:

(A) N
(B) N - 1
(C) N + 1
(D) 2 * N

30. In planar mesh analysis of a circuit with B branches and N nodes, the number of independent mesh equations is:

(A) B - N + 1
(B) B + N - 1
(C) B - N
(D) N - 1

31. A bilateral circuit element is one that:

(A) Conducts in only one direction
(B) Offers identical V-I behavior irrespective of current direction
(C) Has two outputs
(D) Operates only on AC

32. Which of the following is a non-linear circuit element?

(A) Ideal Resistor
(B) PN Junction Diode
(C) Air-core Inductor
(D) Ideal Capacitor

33. A dependent source whose output voltage depends on an input current is called a:

(A) VCVS
(B) VCCS
(C) CCVS
(D) CCCS

34. A dependent source whose output current depends on an input voltage is called a:

(A) VCVS
(B) VCCS
(C) CCVS
(D) CCCS

35. The algebraic sum of voltages around any closed loop in an electrical network is:

(A) Always positive
(B) Always negative
(C) Zero
(D) Equal to total current

36. The unit of electrical energy consumed commercially is kilowatt-hour (kWh), equal to:

(A) 3.6 * 10^5 J
(B) 3.6 * 10^6 J
(C) 1.0 * 10^3 J
(D) 3600 J

37. Work done in moving a unit positive charge from infinity to a point in an electric field is:

(A) Current
(B) Electric potential
(C) Magnetic flux
(D) Capacitance

38. Two identical 12 V batteries connected in parallel provide a terminal open-circuit voltage of:

(A) 24 V
(B) 12 V
(C) 6 V
(D) 0 V

39. Two identical 12 V batteries connected in series aiding provide an open-circuit voltage of:

(A) 12 V
(B) 24 V
(C) 6 V
(D) 0 V

40. The internal resistance of a practical battery causes its terminal voltage on load to:

(A) Increase above EMF
(B) Decrease below EMF
(C) Remain strictly constant
(D) Oscillate

41. In Thevenin's theorem with dependent sources, Rth cannot be found by deactivating sources; instead, one must:

(A) Apply a test voltage/current at load terminals
(B) Short all dependent sources
(C) Open all resistors
(D) Use nodal analysis only

42. Tellegen's theorem is valid for any lumped network provided it obeys:

(A) Ohm's law
(B) Linearity
(C) Kirchhoff's laws (KVL and KCL)
(D) Reciprocity

43. Tellegen's theorem states that the sum of instantaneous powers delivered to all branches is:

(A) Always zero
(B) Always positive
(C) Infinite
(D) Proportional to frequency

44. The Reciprocity theorem is applicable to networks that are:

(A) Linear, bilateral, and contain no dependent sources
(B) Non-linear and unilateral
(C) Active and time-varying
(D) Non-planar

45. In Star-Delta conversion, resistor R_A between nodes 1 and 2 in delta formed from star resistors R1, R2, R3 is:

(A) R1 + R2 + (R1*R2)/R3
(B) R1*R2 / (R1+R2+R3)
(C) (R1+R2)/R3
(D) R1 + R2 + R3

46. In Delta-Star conversion, star resistor R1 connected to node 1 formed from delta resistors Rab, Rbc, Rca is:

(A) (Rab * Rca) / (Rab + Rbc + Rca)
(B) (Rab + Rca) / Rbc
(C) Rab * Rca * Rbc
(D) (Rab * Rbc) / Rca

47. In a DC circuit, an ideal inductor in steady state behaves as a:

(A) Short circuit (0 V drop)
(B) Open circuit (0 A current)
(C) Constant current source
(D) Negative resistance

48. In a DC circuit, an ideal capacitor in steady state behaves as an:

(A) Open circuit (0 A current)
(B) Short circuit
(C) Infinite charge accumulator
(D) Oscillator

49. A 10 V source with 2 Ohm internal resistance delivers maximum power to load RL. The value of maximum power is:

(A) 50 W
(B) 25 W
(C) 12.5 W
(D) 6.25 W

50. Millman's theorem reduces multiple parallel branches of voltage sources V_k with internal resistances R_k into a single equivalent voltage V_eq equal to:

(A) (Sum of V_k * G_k) / (Sum of G_k)
(B) (Sum of V_k) / (Sum of R_k)
(C) (Sum of I_k) * (Sum of R_k)
(D) Zero

51. In mesh analysis, the mutual resistance term R_jk between mesh j and mesh k is always:

(A) Negative when clockwise mesh currents are assumed
(B) Positive always
(C) Zero always
(D) Imaginary

52. In nodal analysis, the off-diagonal conductance term G_ij in the nodal conductance matrix is equal to the:

(A) Negative sum of conductances connected directly between node i and node j
(B) Positive sum of all conductances
(C) Self conductance of node i
(D) Zero

53. The voltage across a 5 Ohm resistor is 20 V. The charge passing through it in 2 minutes is:

(A) 8 Coulombs
(B) 240 Coulombs
(C) 480 Coulombs
(D) 1200 Coulombs

54. A bridge circuit with arms R1=100, R2=200, R3=300, R4=600 Ohms is in a state of:

(A) Balance (no detector current)
(B) Maximum unbalance
(C) Resonance
(D) Oscillation

55. In an unbalanced Wheatstone bridge, the current through the galvanometer can be accurately calculated using:

(A) Thevenin's theorem
(B) Ohm's law alone
(C) Faraday's law
(D) Ampere's law

56. A current of 4 A enters a parallel combination of 3 Ohm and 6 Ohm resistors. Current through the 6 Ohm resistor is:

(A) 2.67 A
(B) 1.33 A
(C) 2.0 A
(D) 0.67 A

57. A voltage divider consists of two resistors R1=4 kOhm and R2=6 kOhm across 50 V. Output across R2 is:

(A) 20 V
(B) 30 V
(C) 25 V
(D) 50 V

58. When a non-ideal voltmeter with resistance Rv is connected across a resistor R to measure voltage, the reading is:

(A) Lower than true voltage (loading effect)
(B) Higher than true voltage
(C) Identical to true voltage
(D) Zero

59. To minimize the loading effect of an ammeter, its internal resistance should be:

(A) As small as possible (approaching zero)
(B) As large as possible
(C) Equal to load resistance
(D) Negative

60. A copper conductor has resistance 10 Ohms at 0 deg C. With temperature coefficient alpha = 0.004 / deg C, resistance at 50 deg C is:

(A) 12 Ohms
(B) 14 Ohms
(C) 20 Ohms
(D) 10.4 Ohms

61. A practical current source consists of an ideal current source I_s in parallel with R_s. Its Thevenin equivalent has:

(A) Vth = I_s * R_s, Rth = R_s in series
(B) Vth = I_s / R_s, Rth = R_s in parallel
(C) Vth = I_s, Rth = 0
(D) Vth = 0, Rth = R_s

62. In a circuit with 4 independent loops and 3 nodes, how many branch currents exist?

(A) 5
(B) 6
(C) 7
(D) 8

63. Compensation theorem states that any branch resistance change delta-R can be replaced by a compensating voltage source of value:

(A) I * delta-R opposing original current
(B) V / delta-R
(C) I^2 * delta-R
(D) Zero

64. A DC circuit has Vth = 24 V and Rth = 8 Ohms. If a load RL = 4 Ohms is connected, the power consumed by RL is:

(A) 16 W
(B) 24 W
(C) 32 W
(D) 18 W

65. In an active circuit containing only dependent sources and resistors, the Thevenin equivalent voltage Vth is:

(A) Zero
(B) Non-zero
(C) Infinite
(D) Indeterminate

66. The dual of a series R-L circuit excited by a voltage source is a:

(A) Parallel G-C circuit excited by a current source
(B) Series R-C circuit
(C) Parallel R-L circuit
(D) Series L-C circuit

67. The dual of mesh analysis is:

(A) Nodal analysis
(B) Loop analysis
(C) Branch analysis
(D) Supermesh analysis

68. In graph theory, the number of branches in a tree (tree branches or twigs) for a graph with N nodes is:

(A) N - 1
(B) N
(C) N + 1
(D) 2 * N

69. In a connected network graph with B branches and N nodes, the number of links (chords) in the co-tree is:

(A) B - N + 1
(B) B + N - 1
(C) N - 1
(D) B - 1

70. The incidence matrix [A] of an electrical network describes the interconnection between:

(A) Branches and nodes
(B) Loops and meshes
(C) Chords and twigs
(D) Cut-sets and ties

71. The rank of the incidence matrix of a connected network with N nodes is:

(A) N - 1
(B) N
(C) N + 1
(D) B

72. The fundamental cut-set matrix [Q_f] has a row for each:

(A) Tree branch (twig)
(B) Link (chord)
(C) Independent loop
(D) Mesh

73. The fundamental tie-set matrix [B_f] has a row for each:

(A) Link (chord)
(B) Tree branch (twig)
(C) Essential node
(D) Reference node

74. If a network satisfies KCL and KVL, the product of branch voltages and branch currents sum to zero. This is:

(A) Tellegen's Theorem
(B) Thevenin's Theorem
(C) Norton's Theorem
(D) Millman's Theorem

75. Maximum power transfer theorem for a complex source impedance Zth delivering to load ZL = RL + jXL requires:

(A) ZL = Zth* (complex conjugate)
(B) ZL = Zth
(C) RL = 0
(D) XL = Xth

76. In a network containing a current-controlled current source (CCCS) with gain beta, the nodal admittance matrix:

(A) Loses symmetry (Y_ij != Y_ji)
(B) Remains strictly symmetric
(C) Becomes diagonal
(D) Has zero determinant

77. A supernode equation combines KCL across Gaussian surface enclosing the voltage source with:

(A) A constraint equation linking the node voltages via the source voltage
(B) A KVL around outer loop
(C) Thevenin impedance
(D) Norton admittance

78. A DC network contains an independent source V_s and a VCVS mu*V_x. If V_s is doubled, all branch voltages:

(A) Double (homogeneity property of linear systems)
(B) Quadruple
(C) Remain unchanged
(D) Halve

79. In Modified Nodal Analysis (MNA), independent voltage sources are incorporated by:

(A) Adding branch currents as extra auxiliary variables in the matrix system
(B) Converting them to Norton equivalents
(C) Neglecting them
(D) Replacing them with capacitors

80. When determining Rth for a circuit with dependent sources using a 1 V test source, if test current I_test = -0.5 A, Rth is:

(A) -2 Ohms (active network exhibiting negative dynamic resistance)
(B) +2 Ohms
(C) 0.5 Ohms
(D) Infinite

81. A negative Thevenin resistance Rth in an active circuit signifies that the circuit:

(A) Can supply net energy to an external port rather than dissipating it
(B) Is physically impossible
(C) Violates conservation of energy
(D) Is purely capacitive

82. In a bridge T-attenuator, source and load resistances are R0. Characteristic resistance of the pad is designed to:

(A) Match R0 to provide impedance matching without reflection
(B) Provide infinite attenuation
(C) Maximize distortion
(D) Eliminate DC flow

83. The driving-point input impedance of a terminated two-port network depends on:

(A) Network parameters and load impedance
(B) Load impedance only
(C) Source voltage only
(D) Frequency only

84. In a symmetrical lattice network with series arm Za and diagonal arm Zb, the characteristic impedance Z0 is:

(A) sqrt(Za * Zb)
(B) Za + Zb
(C) Za * Zb
(D) sqrt(Za / Zb)

85. Bartlett's Bisection Theorem allows the determination of equivalent lattice arms for a:

(A) Symmetrical two-port network
(B) Non-linear network
(C) Unilateral amplifier
(D) Distributed line

86. A gyrator is a linear, lossless, non-reciprocal two-port network whose V-I characteristic relates:

(A) V1 = -r*I2 and V2 = r*I1
(B) V1 = r*I1 and V2 = r*I2
(C) V1 = V2
(D) I1 = -I2

87. When a capacitor C is connected across port 2 of an ideal gyrator with gyration resistance r, port 1 behaves as an:

(A) Inductor of value L = C * r^2
(B) Capacitor of value C / r^2
(C) Ideal resistor r
(D) Negative capacitor

88. A Negative Impedance Converter (NIC) terminates in an impedance Z_L. The input impedance seen at the port is:

(A) -k * Z_L
(B) +k * Z_L
(C) 1 / Z_L
(D) Zero

89. In a network graph, the orthogonality relation between fundamental cut-set matrix [Q_f] and tie-set matrix [B_f] is:

(A) [Q_f] * [B_f]^T = [0]
(B) [Q_f] * [B_f] = [I]
(C) [Q_f] + [B_f] = [0]
(D) [Q_f] = [B_f]

90. The relation connecting branch voltages [V_b] and node voltages [V_n] via incidence matrix [A] is:

(A) [V_b] = [A]^T * [V_n]
(B) [V_b] = [A] * [V_n]
(C) [V_n] = [A] * [V_b]
(D) [V_b] = [A]^-1 * [V_n]

91. The relation connecting branch currents [I_b] and KCL at all non-reference nodes is expressed as:

(A) * [I_b] = [0]
(B) [A]^T * [I_b] = [0]
(C) [I_b] = [A] * [I_n]
(D) [A] + [I_b] = [0]

92. In equilibrium node analysis, the node admittance matrix is formulated using graph matrices as:

(A) [Y_node] = [A] * [Y_branch] * [A]^T
(B) [Y_node] = [A]^T * [Y_branch] * [A]
(C) [Y_node] = [A] * [Y_branch]
(D) [Y_node] = [Y_branch] * [A]

93. In equilibrium loop analysis, the loop impedance matrix is formulated using tie-set matrix [B] as:

(A) [Z_loop] = [B] * [Z_branch] * [B]^T
(B) [Z_loop] = [B]^T * [Z_branch] * [B]
(C) [Z_loop] = [B] * [Z_branch]
(D) [Z_loop] = [Z_branch] * [B]

94. If a DC circuit has a non-unique solution in nodal analysis, the node admittance matrix [Y_n]:

(A) Is singular (det[Y_n] = 0)
(B) Has positive determinant
(C) Is an identity matrix
(D) Is symmetric positive definite

95. A resistor network has two terminals A-B. If connecting 10 V causes 2 A, and adding 5 Ohms in series causes 1 A, source Vth is:

(A) 10 V and Rth = 5 Ohms
(B) 20 V and Rth = 5 Ohms
(C) 15 V and Rth = 2.5 Ohms
(D) 5 V and Rth = 2 Ohms

96. A current source I_s is in parallel with R1 and R2. The current through R1 is given by the current divider formula:

(A) I_s * R2 / (R1 + R2)
(B) I_s * R1 / (R1 + R2)
(C) I_s * (R1 + R2) / R2
(D) I_s * R1 * R2

97. A non-linear resistor has V-I relation V = 2*I + 3*I^2. For small variations around quiescent point I_0 = 2 A, dynamic resistance is:

(A) 14 Ohms
(B) 8 Ohms
(C) 10 Ohms
(D) 2 Ohms

98. An infinite ladder network consists of repeating series resistors R and shunt resistors R. The input resistance R_in is:

(A) R * (1 + sqrt(5)) / 2
(B) R * sqrt(2)
(C) 2 * R
(D) R / 2

99. In an infinite ladder network of identical 1 Ohm resistors, the input resistance satisfies R_in = 1 + (1*R_in)/(1+R_in), giving:

(A) (1 + sqrt(5)) / 2 = 1.618 Ohms
(B) 2.0 Ohms
(C) 1.414 Ohms
(D) 1.0 Ohm

100. In a bridge network with delta-star equivalent, the total power dissipated calculated via equivalent circuit is:

(A) Exactly identical to total power dissipated in the original delta network
(B) Greater in delta
(C) Smaller in star
(D) Dependent on ground node

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. State and prove Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) from fundamental physical conservation principles.

Q2. State Thevenin's Theorem and provide a rigorous step-by-step mathematical derivation of its equivalent parameters.

Q3. State Norton's Theorem, compare it with Thevenin's Theorem, and demonstrate source transformation.

Q4. State and prove the Maximum Power Transfer Theorem for DC resistive circuits, and determine the efficiency at maximum power.

Q5. Derive the mathematical transformation formulas for Delta-to-Star and Star-to-Delta resistor conversions.

Q6. Detail Nodal Analysis methodology, including matrix formulation and the treatment of supernodes.

Q7. Explain Mesh Analysis methodology, matrix formulation, and handling of supermeshes containing current sources.

Q8. State and prove Tellegen's Theorem, explaining why it holds for linear, non-linear, time-varying, and active circuits.

Q9. State and explain the Superposition Theorem, detailing its scope, proof, and why it cannot be directly applied to power.

Q10. Explain Millman's Theorem and demonstrate how it simplifies parallel generator branches into an equivalent source.

Q11. State the Reciprocity Theorem, define its validity conditions, and illustrate with a two-port resistive network.

Q12. State the Compensation Theorem and explain its application in sensitivity analysis and bridge circuits.

Q13. Explain the concept of Duality in electrical circuits, listing dual pairs and steps to construct a dual network graph.

Q14. Formulate the fundamental matrices in Network Graph Theory: Incidence matrix [A], Tie-set matrix [B], and Cut-set matrix [Q].

Q15. Solve a bridge circuit: Calculate Thevenin equivalent across load terminals of an unbalanced Wheatstone bridge.

Q16. Explain how to determine Thevenin resistance in active networks containing dependent sources using test sources.

Q17. Describe the Loading Effect in voltmeter and ammeter measurements, deriving correction factors for true values.

Q18. Explain Temperature Coefficient of Resistance, deriving formulas for resistance variation and temperature rise in conductors.

Q19. Solve an active circuit problem containing a Voltage-Controlled Voltage Source (VCVS) using Nodal Analysis.

Q20. Analyze an Infinite Ladder Resistive Network, deriving the exact formula for driving-point input resistance.

Q21. Explain the concept of Supermesh in detail with a complete worked mathematical example.

Q22. Explain Modified Nodal Analysis (MNA) and its fundamental role in SPICE circuit simulators.

Q23. Discuss Non-Linear DC Circuit Analysis using the Load-Line Method and Newton-Raphson numerical iteration.

Q24. State and prove Bartlett's Bisection Theorem for symmetrical two-port networks.

Q25. Explain Star-Delta equivalence in terms of multi-terminal networks and driving-point conductance.

Q26. Derive the conditions for zero current in the galvanometer of an Anderson Bridge and a Kelvin Double Bridge.

Q27. Explain the principle and operation of a Negative Impedance Converter (NIC) and its circuit implementation using an Op-Amp.

Q28. Explain the Gyrator circuit, its impedance inversion properties, and how it synthesizes inductors using capacitors.

Q29. Demonstrate Source Shifting (Blakesley's Theorem) for voltage sources and current sources.

Q30. Analyze Power Distribution in a 3-Wire DC Distribution System, comparing copper efficiency with a 2-Wire system.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[B]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[C]	**Q62:**	[B]	**Q82:**	[A]
Q3:	[B]	**Q23:**	[C]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[C]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[C]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[B]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[B]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[B]	**Q28:**	[B]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[B]	**Q29:**	[B]	**Q49:**	[C]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[B]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[B]	**Q31:**	[B]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[C]	**Q33:**	[C]	**Q53:**	[C]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[B]	**Q34:**	[B]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[C]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[B]	**Q56:**	[B]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[B]	**Q37:**	[B]	**Q57:**	[B]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[B]	**Q38:**	[B]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[B]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

voltage source V_L is active, driving current into terminal A. Since the internal network is purely resistive, the input impedance seen by V_L is Rth. The current flowing out of terminal A is -V_L / Rth.

- Combining both contributions via superposition:

I_L = I_sc - V_L / Rth = I_sc - (I_L * RL) / Rth.

- Under open circuit (RL -> infinity, I_L = 0):

0 = I_sc - V_oc / Rth => V_oc = I_sc * Rth. Define Vth = V_oc.

- Substitute I_sc = Vth / Rth back into the current equation:

I_L = Vth / Rth - (I_L * RL) / Rth => I_L * (1 + RL / Rth) = Vth / Rth => I_L * (Rth + RL) = Vth.

- Thus: I_L = Vth / (Rth + RL), which represents a series loop of source Vth, internal resistance Rth, and load RL.

>>Solution to Question 8: Modern Power Transformers are designed to operate with extremely high efficiency at maximum power.

derivative to zero:
dP_L / dRL = Vth^2 * [(Rth + RL)^2 * 1 - RL * 2 * (Rth + RL)] / (Rth + RL)^4 = 0.
- Simplify numerator:
(Rth + RL) - 2 * RL = 0 => Rth - RL = 0 => RL = Rth.
- Verify second derivative: d^2 P_L / dRL^2 at RL = Rth is negative (-Vth^2 / 8*Rth^3 < 0), confirming a global maximum.
3. Value of Maximum Power:
P_max = (Vth^2 * Rth) / (Rth + Rth)^2 = (Vth^2 * Rth) / (4 * Rth^2) = Vth^2 / (4 * Rth).
4. Efficiency Calculation:
- Total power generated by source: P_total = Vth * I_L = Vth * [Vth / (2*Rth)] = Vth^2 / (2*Rth).
- Efficiency eta = P_L / P_total = [Vth^2 / (4*Rth)] / [Vth^2 / (2*Rth)] = 1/2 = 50%.
- At maximum power transfer, 50% of the energy is lost internally inside source resistance Rth as heat.

>>Solution Steps: 1. Nodal Analysis: Apply KCL at node A. 2. Mesh Analysis: Apply KVL to the two loops. 3. Source Transformation: Convert the current source to a voltage source. 4. Thevenin's Theorem: Find Vth and Rth. 5. Norton's Theorem: Find In and Rn. 6. Star-Delta Transformation: Use formulas for conversions.

- Express branch currents in terms of node voltages: I_ij = (Vi - Vj) / R_ij.
2. Matrix Formulation ([G] * [V] = [I]):
- [G] is an (N-1) x (N-1) conductance matrix:
* Diagonal entries G_ii: Self-conductance (sum of conductances connected directly to node i).
* Off-diagonal entries G_ij: Mutual conductance (-sum of conductances directly connecting node i and j).
- [V] is the vector of node voltages [V1, V2, ..., V_{N-1}]^T.
- [I] is the vector of source currents entering node i.
3. Treatment of Supernodes:
- Occurs when an ideal independent or dependent voltage source is connected between two non-reference nodes (e.g., between node j and node k).
- Because the current through an ideal voltage source is arbitrary and cannot be determined by Ohm's law, enclose the voltage source and its two nodes in a generalized boundary (Supernode).
- Write one KCL equation for the supernode surface enclosing both nodes j and k.
- Supplement with an auxiliary constraint equation relating node voltages: Vj - Vk = V_source.

>>Solution to Question 8: Thevenin's Theorem. extension of mesh analysis for linear circuits containing dependent sources.

- Let [A] be the (N-1) x B reduced incidence matrix of the network graph.
- By KCL, the branch currents satisfy: [A] * [i_b] = [0].
- By KVL, the branch voltages are related to node voltages [v_n] by: [v_b] = [A]^T * [v_n].
- Evaluate total instantaneous power:
Sum_{k=1}^B (v_k * i_k) = [v_b]^T * [i_b].
- Substitute [v_b]^T = ([A]^T * [v_n])^T = [v_n]^T * [A]:
Sum_{k=1}^B (v_k * i_k) = [v_n]^T * ([A] * [i_b]).
- Since [A] * [i_b] = [0] by KCL:
Sum_{k=1}^B (v_k * i_k) = [v_n]^T * [0] = 0.
3. Universality of Tellegen's Theorem:
- The proof relies exclusively on network topology (incidence matrix [A]), KCL, and KVL.
- It makes NO assumption about branch constitutive relationships (Ohm's law).
- Therefore, it is valid for linear, non-linear, passive, active, time-invariant, and
time-varying networks.

>>Solution Step-by-Step Question Bank Solutions: Tellegen's Theorem | Network Topology | Power Conservation

- R_eq = 1 / (Sum_{k=1}^n G_k).

2. Analytical Derivation:

- Let terminals A and B be open-circuited. Choose terminal B as the reference node (0 V) and let the potential of terminal A be V_AB.
- Apply KCL at node A:

Sum_{k=1}^n I_k = 0 => Sum_{k=1}^n (V_AB - V_k) / R_k = 0.

- Expand summation:

V_AB * Sum_{k=1}^n (1 / R_k) - Sum_{k=1}^n (V_k / R_k) = 0.

- Since G_k = 1 / R_k:

V_AB * Sum(G_k) = Sum(V_k * G_k) => V_AB = [Sum(V_k * G_k)] / [Sum(G_k)].

- This terminal open-circuit voltage is V_eq.
- To find internal resistance R_eq, deactivate all independent sources (V_k = 0). The parallel combination of all conductances yields:

G_eq = Sum_{k=1}^n G_k => R_eq = 1 / G_eq.

>> Solution to Question 12: Reciprocity Theorem and its validity conditions and illustrate with a two-port resistive network.

- This can be viewed as the original resistance R in series with an added potential drop V_c = I' * delta-R.
- For small variations, V_c approx I * delta-R.

3. Practical Applications:
- Bridge Sensitivity: Used to determine the unbalance current in Wheatstone and Kelvin bridges caused by small sensor resistance changes (strain gauges, RTDs).
- Sensitivity Analysis: Allows rapid evaluation of power grid and circuit tolerance deviations without re-solving the full nodal matrix.

>>**Solution to Question 14:** Derive the formula for Dual in electric Networks. Use Thevenin's theorem to find the current through a load resistor in a bridge circuit.

- Dimension: (B - N + 1) x B. Each link forms exactly one fundamental loop with tree branches.
- Elements b_ij = +1 if branch j orientation matches loop direction; -1 if opposite; 0 if not in loop.
- KVL formulation: [B_f] * [V_b] = [0].

4. Fundamental Cut-Set Matrix [Q_f]:
- Dimension: (N - 1) x B. Each twig defines a unique fundamental cut-set with links.
- Elements q_ij = +1 if branch j matches cut-set orientation; -1 if opposite; 0 otherwise.
- KCL formulation: [Q_f] * [I_b] = [0].

5. Orthogonality Property: [B_f] * [Q_f]^T = [0] and [A] * [B_f]^T = [0].

>>Solution Question 6: a node determine Calculate The resistance active networks contains fundamental Wheatstone bridge.

* Option A: Apply test voltage source V_test (typically 1 V).
* Option B: Apply test current source I_test (typically 1 A).
- Step 3: Solve the circuit using nodal or mesh analysis to compute the current I_test leaving V_test (or voltage V_test across I_test).
- Step 4: The Thevenin resistance is:
Rth = V_test / I_test.
3. Physical Significance of Negative Rth:
- In active circuits with positive feedback, I_test can flow into the positive terminal of V_test (I_test < 0).
- This results in a negative driving-point resistance (Rth < 0).
- An active negative resistance port acts as an energy source rather than a dissipator, capable of driving sustained oscillations or circuit instability.

>>Solution Question 15: The Temperature Coefficient of Resistance denotes the change in resistance per degree change in temperature relative to the resistance value at a specific reference temperature, usually 20°C. It is used to calculate how much the resistance of a material will increase or decrease as the temperature rises or falls from that reference point. Positive Temperature Coefficient (PTC): Most metals, such as copper and aluminum, have a positive temperature coefficient, meaning their resistance increases as the temperature increases. This is because the increased thermal energy causes more collisions between free electrons and the metal's atoms, hindering the flow of current. Negative Temperature Coefficient (NTC): Some materials, like semiconductors, have a negative temperature coefficient, meaning their resistance decreases as the temperature increases. This is because the increased thermal energy provides more free electrons, which can carry more current. Zero Temperature Coefficient: Some materials, like certain alloys, have a zero temperature coefficient, meaning their resistance remains relatively constant over a wide range of temperatures. Formula: The change in resistance can be calculated using the formula: ΔR = R0 * α * ΔT, where ΔR is the change in resistance, R0 is the resistance at the reference temperature, α is the temperature coefficient of resistance, and ΔT is the change in temperature. Applications: Temperature coefficients are important in various applications, such as designing resistors for specific temperature ranges, compensating for temperature variations in circuits, and developing temperature sensors.

alpha_2 = alpha_1 / [1 + alpha_1 * (T2 - T1)].

- Inferred Zero Resistance Temperature (T0):

For copper, extrapolating the linear resistance curve to zero gives T0 = -234.5 deg C.

Ratio formula: R(T2) / R(T1) = (234.5 + T2) / (234.5 + T1).

4. Measurement of Machine Winding Temperature Rise:

- Electrical machine thermal rise under load is measured by comparing cold winding resistance R1 at ambient T1 with hot winding resistance R2:

T2 = [R2 / R1] * (234.5 + T1) - 234.5.

>>Solution Question 20 Analyze active ladder resistive network using Thevenin theorem Solve for VCVS using Mesh Analysis.

- Therefore, the input resistance looking into the second section is identically equal to the total driving-point resistance R_in seen at the main input terminals.

3. Equivalent Circuit Equation:

- The total circuit reduces to series resistor R1 in series with the parallel combination of shunt resistor R2 and R_in:

R_in = R1 + [R2 * R_in] / (R2 + R_in).

4. Quadratic Solution:

- Clear denominator by multiplying through by (R2 + R_in):

R_in * (R2 + R_in) = R1 * (R2 + R_in) + R2 * R_in.

R_in^2 + R2 * R_in = R1 * R2 + R1 * R_in + R2 * R_in.

R_in^2 - R1 * R_in - R1 * R2 = 0.

- Applying the quadratic formula (taking the positive root since resistance must be positive):

R_in = [R1 + sqrt(R1^2 + 4 * R1 * R2)] / 2.

5. Special Case (R1 = R2 = R):

- R_in = [R + sqrt(R^2 + 4 * R^2)] / 2 = R * [1 + sqrt(5)] / 2 = 1.61803 * R (Golden Ratio phi * R).

>>Solution to Question 22 in Modified Nodal Analysis (MNA) deal with fundamental worked SPICE circuit examples.

admittance matrix singular.
2. Formulation of MNA:
- Partition circuit unknowns into:
(a) N-1 Node voltages [v].
(b) Currents through independent and dependent voltage sources [i_v].
- The system is cast into block matrix form:
[G B] [v] = [i_s]
[C D] [i_v] [e_s]
- Matrix [G]: Standard nodal conductance matrix formed by passive resistors/conductances.
- Matrix [B]: Topological incidence matrix routing voltage source currents into nodes.
- Matrix [C]: Links node voltages to voltage source branch equations (for independent sources, C = B^T).
- Matrix [D]: Accounts for dependent voltage source couplings (zeros for independent sources).
- Vector [i_s]: Independent current sources entering nodes.
- Vector [e_s]: Values of independent voltage sources.
3. Algorithmic Advantage:
- Guaranteed non-singular matrix for physically realizable circuits, easily assembled via element 'stamps' in computer code (SPICE).

>>Solution to Question 22 Nonlinear DC Circuit Analysis Using the Newton-Raphson numerical iteration.

- Series lattice arm impedance: Z_a = Z_oc/2 or Z_sc/2 depending on open/short conditions.
- Specifically:
Z_a is the input impedance of one half-network with all bisected connecting lines OPEN-CIRCUITED (even excitation).
Z_b is the input impedance of one half-network with all bisected connecting lines SHORT-CIRCUITED (odd excitation).
2. Proof Outline via Even and Odd Mode Decomposition:
- Any arbitrary excitation at port 1 (V1) and port 2 (V2) can be decomposed into:
(a) Even mode: V_e = (V1 + V2)/2 applied to both ports (no current crosses symmetry plane -> Open Circuit bisection).
(b) Odd mode: V_o = (V1 - V2)/2 applied anti-symmetrically (voltage is zero at symmetry plane -> Short Circuit bisection).
- Input impedances in even and odd modes correspond directly to lattice arms Z_a and Z_b.
3. Utility:
- Drastically simplifies the design and analysis of symmetric bridge-T filters, crystal filters, and attenuators.

>>Solution to Question 25: Star-Delta conversion for zero current in the cross-arm of a balanced Wheatstone Bridge and Kelvin Double Bridge.

R = (P / Q) * S.

- Lead resistance 'r' has zero effect on the measured resistance R.

2. Standard Wheatstone Null Condition:

- Ratio of adjacent arms: R1 / R2 = R3 / R4 => R1 * R4 = R2 * R3.

>>Solution to Question 28 Explain the Gyrator and its application in realization of Inductors and Transformers. Can it be (NO) used as circuit element in an Op-Amp.

3. Active Inductor Synthesis:
- If port 2 is terminated with a capacitor: Z_L = 1 / (s * C).
- Input impedance: Z_in = r^2 / [1 / (s * C)] = s * (r^2 * C).
- This matches the impedance of an ideal inductor: Z_in = s * L_eq, where:
L_eq = r^2 * C.
- Eliminates bulky, lossy physical magnetic coils in Integrated Circuit (IC) design.

>>Solution Store Question Bank Power Point Slides in PDF Video Tutorial for Students and more efficiency with a 2-Wire system.

half the outer (a_N = a2 / 2).
P_loss = 2 * I2^2 * R2 = 2 * (I1 / 2)^2 * (rho * L / a2) = 0.5 * (I1^2 * rho * L / a2).
Equating losses: a2 = a1 / 4.
Volume of copper:
Vol_3 = (a2 + a2 + 0.5*a2) * L = 2.5 * a2 * L = 2.5 * (a1 / 4) * L = 0.3125 * (2 * a1 * L) =
0.3125 * Vol_2 (31.25%).
- Copper Savings: The 3-wire DC distribution system achieves a massive 68.75% reduction in copper requirement compared to a standard 2-wire system.
